MILITARY SPECIFICATION

MICROCIRCUITS, DIGITAL, BIPOLAR, ADVANCED SCHOTTKY TTL, PARITY CHECKER, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, advanced schottky TTL, parity checker microcircuit. Two product assurance classes and a choice of case outlines and lead finishes are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	9-bit odd/even parity generator/checker

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outlines. The case outlines shall be designated as follows:

Outline letter	Case outline (MIL-M-38510, appendix C)
A	F-1 (14-lead, 1/4" x 1/4"), flat package
B	F-3 (14-lead, 3/16" x 1/4"), flat package
C	D-1 (14-lead, 1/4" x 3/4"), dual-in-line package
D	F-2 (14-lead, 1/4" x 3/8"), flat package
2	C-2 (20 terminal, .350" x .350"), square chip carrier package

1.3 Absolute maximum ratings.

Supply voltage range	-0.5 V dc to +7.0 V dc
Input voltage range	-1.2 V dc at -18 mA to +7.0 V dc
Storage temperature range	-65°C to +150°C
Maximum power dissipation (P_D) per device 1/	220 mW
Lead temperature (soldering, 10 seconds)	300°C
Thermal resistance, junction-to-case (θ_{JC}):	
Cases A, B, C, and D	(See MIL-M-38510, appendix C)
Case 2	60°C/W 2/
Junction temperature (T_J) 3/	+175°C

1/ Must withstand the added P_D due to short circuit test (e.g., I_{OS}).
2/ When a thermal resistance value is included in MIL-M-38510, appendix C, it shall supersede the value stated herein.
3/ Maximum junction temperature shall not be exceeded except for allowable short duration burn-in screening conditions per method 5004 of MIL-STD-883.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.4 Recommended operating conditions.

Supply voltage (V_{CC}) - - - - - - - - - - - - - - - -	4.5 V dc to 5.5 V dc
Minimum high level input voltage (V_{IH}) - - - - -	2.0 V dc
Maximum low level input voltage (V_{IL})- - - - - -	0.8 V dc
Case operating temperature range (T_C)- - - - - -	-55°C to +125°C

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510, and herein.

3.2.1 Logic diagrams and terminal connections. The logic diagrams and terminal connections shall be as specified on figures 1 and 2.

3.2.2 Truth tables. The truth tables shall be as specified on figure 3.

3.2.3 Schematic circuits. Schematic circuits shall be submitted to the preparing activity prior to inclusion of a manufacturer's device in the specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturers' schematics shall be maintained by the agent activity and will be available upon request.

3.2.4 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.4).

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and apply over the full recommended case operating temperature range, unless otherwise specified.

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TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions $-55°C \leq T_C \leq +125°C$	Device type	Limits		Unit
				Min	Max	
High-level output voltage	V_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.8 V V_{IH} = 2.0 V, I_{OH} = -1 mA	01	2.5		V
Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IL} = 0.8 V V_{IH} = 2.0 V, I_{OL} = 20 mA	01		0.5	V
Input clamp voltage	V_{IC}	V_{CC} = 4.5 V, I_{IN} = -18 mA T_C = +25°C	01		-1.2	V
Low-level input current	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.5 V	01	-.03	-.60	mA
High-level input current	I_{IH1}	V_{CC} = 5.5 V, V_{IH} = 2.7 V	01		20	µA
	I_{IH2}	V_{CC} = 5.5 V, V_{IH} = 7.0 V	01		100	µA
Short circuit output current 1/	I_{OS}	V_{CC} = 5.5 V, V_{OS} = 0.0 V	01	-60	-150	mA
Output drive	I_{OD}	V_{CC} = 4.5 V, V_{OUT} = 2.5 V, V_{IN} = 5.5 V	01	60		mA
Supply current	I_{CC}	V_{CC} = 5.5 V, V_{IN} = 0.0 V	01		40	mA
Propagation delay time, low-to-high level		V_{CC} = 5.0 V (see figure 5) C_L = 50 pF ±10% R_L = 500Ω				
Input to output	t_{PLH}		01	1.0	20.0	ns
Propagation delay time, high-to-low level		V_{CC} = 5.0 V (see figure 5)				
Input to output	t_{PHL}		01	1.0	21.0	ns

1/ Not more than one output should be shorted at a time.

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 11 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) shall be as specified in MIL-M-38510.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1*,2,3,7, 8,9,10,11	1*,2,3,7, 9
Group A test requirements (method 5005)	1,2,3,7, 8,9,10,11	1,2,3,7, 8,9
Group B test requirements (method 5005) subgroup 5	1,2,3,9, 10,11	N/A
Group C end-point electrical parameters (method 5005)	N/A	1,2,3
Additional electrical subgroups for group C periodic inspections	N/A	
Group D end-point electrical parameters (method 5005)	1,2,3	1,2,3

*PDA applies to subgroup 1 (see 4.2c).

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Device type 01
Cases A, B, C, and D

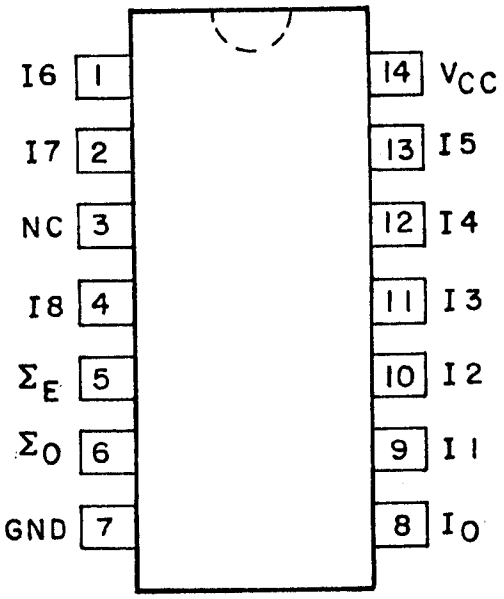


FIGURE 1. Terminal connections.

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Device type 01

Case 2

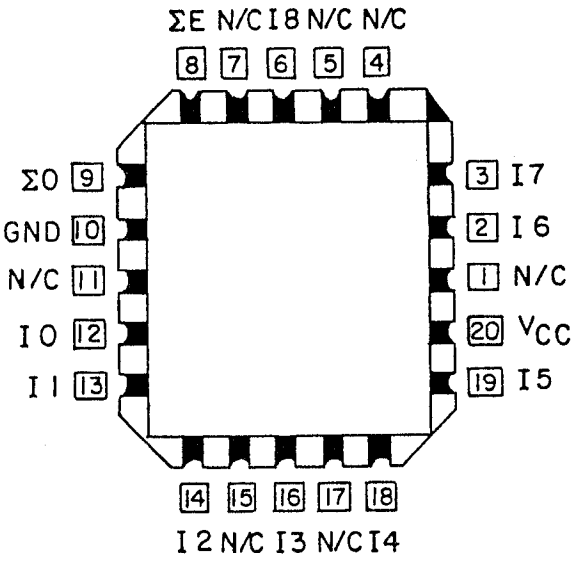


FIGURE 1. Terminal connections (top view) - Continued.

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Device type 01

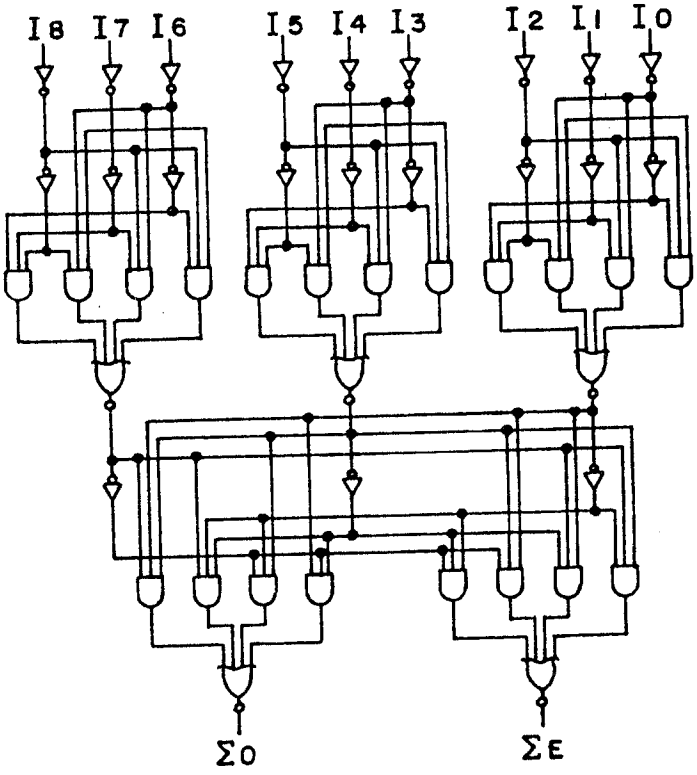


FIGURE 2. Logic diagrams.

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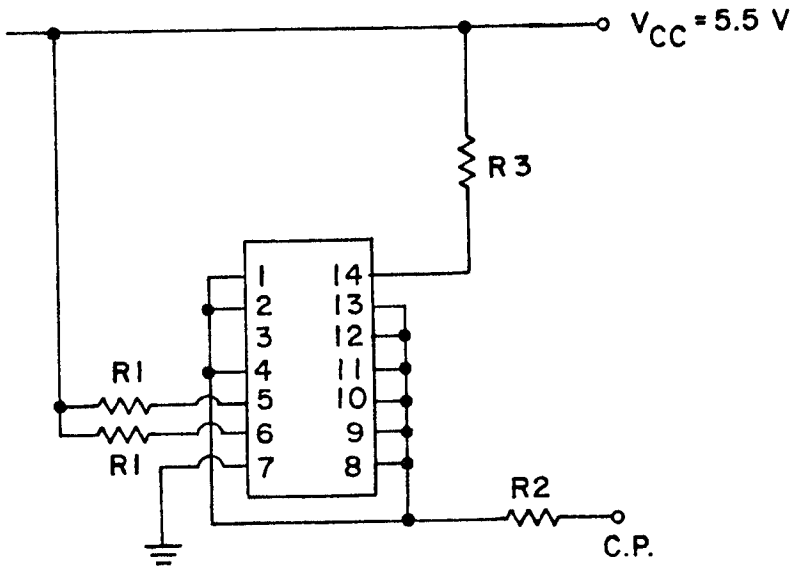
Device type 01

Number of inputs I_0 - I_8 that are high	Outputs	
	Σ Even	Σ Odd
0, 2, 4, 6, 8	H	L
1, 3, 5, 7, 9	L	H

H = High voltage level.
L = Low voltage level.

FIGURE 3. Truth tables.

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NOTES:

1. C.P. = 100 kHz ±50% square wave; duty cycle = 50 ±15%; V_{IL} = -0.5 V minimum to 0.8 V maximum; V_{IN} = 2.0 V minimum to 5.5 V maximum.
2. R_1 = 240 Ω ±5%, R_2 = 68 Ω ±5%.
3. R_3 and V_{CC} shall be chosen to insure 5.5 V minimum is present at device V_{CC} terminal.

FIGURE 4. Burn-in and life test circuit.

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Device type 01

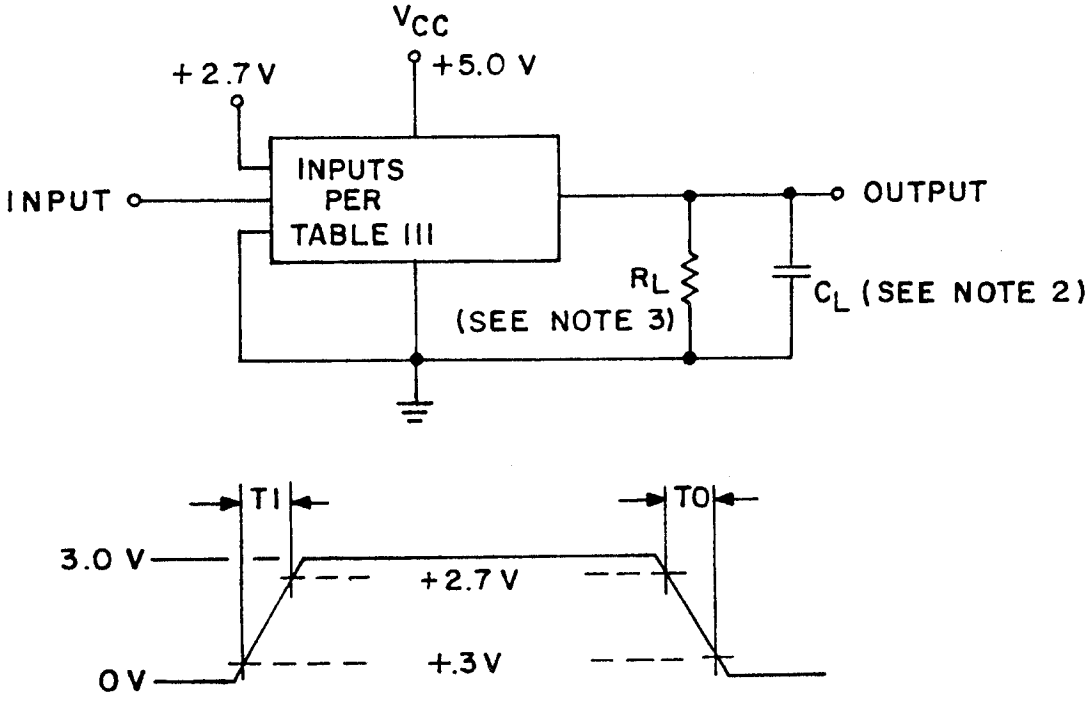
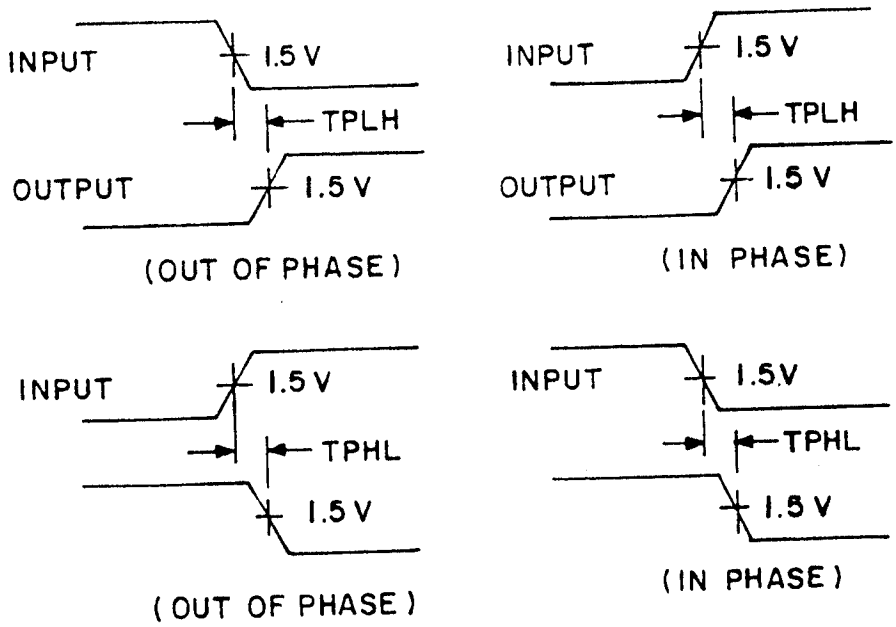


FIGURE 5. Switching time waveforms.

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Device type 01



NOTES:
1. Input pulse has the following characteristics: $T_1 = T_0 \leq 2.5$ ns, PRR $\leq$ 1 MHz, $Z_{OUT} = 50\Omega$.
2. $C_L = 50$ pF ±10% including scope probe, wiring and stray capacitance without package in test fixture.
3. $R_L = 500\Omega$ ±5%.

FIGURE 5. Switching time waveforms - Continued.

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TABLE III. Group A inspection for device type 01.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits Min	Limits Max	Unit
				2	3	4	6	8	9	10	12	13	14	16	18	19	20				
				I6	I7	NC	I8	ΣE	ΣO	GND	IO	I1	I2	I3	I4	I5	V_{CC}				
1 T_C = +25°C	V_{OL}	3007	1	0.8 V	0.8 V		0.8 V	20 mA	20 mA	GND	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	4.5 V	ΣO		0.5	V
			2	2.0 V	2 .0 V		2.0 V			"	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	"	ΣE		0.5	"
	V_{OH}	3006	3	0.8 V	0.8 V		2.0 V	-1 mA	-1 mA	"	0.8 V	0.8 V	2.0 V	0.8 V	0.8 V	2.0 V	"	ΣO	2.5		"
			4	0.8 V	0.8 V		0.8 V			"	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	0.8 V	"	ΣE	2.5		"
	V_{IC}		5	-18 mA	-18 mA		-18 mA			"	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	-18 mA	"	I6		-1.2	"
			6							"							"	I7		"	"
			7							"							"	I8		"	"
			8							"							"	IO		"	"
			9							"							"	I1		"	"
			10							"							"	I2		"	"
			11							"							"	I3		"	"
			12							"							"	I4		"	"
			13							"							"	I5		"	"
	I_{IH1}	3010	14	2.7 V	2.7 V		2.7 V			"	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	2.7 V	5.5 V	I6		20	µA
			15							"							"	I7		"	"
			16							"							"	I8		"	"
			17							"							"	IO		"	"
			18							"							"	I1		"	"
			19							"							"	I2		"	"
			20							"							"	I3		"	"
			21							"							"	I4		"	"
			22							"							"	I5		"	"
	I_{IH2}		23	7.0 V	7.0 V		7.0 V			"	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	7.0 V	"	I6		100	"
			24							"							"	I7		"	"
			25							"							"	I8		"	"
			26							"							"	IO		"	"
			27							"							"	I1		"	"
			28							"							"	I2		"	"
			29							"							"	I3		"	"
			30							"							"	I4		"	"
			31							"							"	I5		"	"
	I_{IL}	3009	32	0.5 V	0.5 V		0.5 V			"	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	0.5 V	"	I6	2/	2/	2/
			33							"							"	I7	"	"	"
			34							"							"	I8	"	"	"
			35							"							"	IO	"	"	"
			36							"							"	I1	"	"	"
			37							"							"	I2	"	"	"
			38							"							"	I3	"	"	"
			39							"							"	I4	"	"	"
			40							"							"	I5	"	"	"
	I_{OS}	3011	41	0.0 V	0.0 V		5.5 V		0.0 V	"	0.0 V	0.0 V	5.5 V	0.0 V	0.0 V	5.5 V	"	ΣO	-60	-150	mA
			42	"	"		0.0 V	0.0 V		"	"	"	0.0 V	"	"	0.0 V	"	ΣE	-60	-150	"
	I_{OD}		43	"	"		5.5 V	2.5 V		"	"	"	5.5 V	"	"	5.5 V	4.5 V	ΣE	60		"
			44	"	"		0.0 V		2.5 V	"	"	"	0.0 V	"	"	0.0 V	4.5 V	ΣO	60		"
	I_{CC}	3005	45	"	"		0.0 V			"	"	"	0.0 V	"	"	0.0 V	5.5 V	V_{CC}		40	"
2	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = +125°C and V_{IC} tests are omitted.																				
3	Same tests, terminal conditions, and limits as for subgroup 1, except T_C = -55°C and V_{IC} tests are omitted.																				

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	16	17	NC	18	ΣE	ΣO	GND	10	11	12	13	14	15	V_{CC}				
7 T_C = +25°C	Functional tests 4/	3014	46	B	B		B	H	L	GND	B	B	B	B	B	B	3/	All outputs			
			47	"	B		A	L	H	"	"	B	A	"	B	A	"	"			
			48	"	A		B	L	L	"	"	A	B	"	A	B	"	"			
			49	"	B		B	H	H	"	"	B	A	"	B	A	"	"			
			50	"	B		A	L	H	"	"	A	B	"	A	B	"	"			
			51	"	A		B	H	L	"	"	B	A	"	B	A	"	"			
			52	"	B		A	L	H	"	"	A	B	"	A	B	"	"			
			53	"	B		B	H	L	"	"	B	A	"	B	A	"	"			
			54	"	A		A	L	H	"	"	A	B	"	A	B	"	"			
			55	"	B		B	H	L	"	"	B	A	"	B	A	"	"			
			56	"	B		A	L	H	"	"	A	B	"	A	B	"	"			
			57	"	A		B	H	L	"	"	B	A	"	B	A	"	"			
			58	"	B		A	L	H	"	"	A	B	"	A	B	"	"			
			59	"	B		B	H	L	"	"	B	A	"	B	A	"	"			
			60	"	A		B	L	H	"	"	A	B	"	A	B	"	"			
			61	"	B		A	H	L	"	"	B	A	"	B	A	"	"			
8	Same tests, terminal conditions, and limits as for subgroup 7, except T_C = +125°C and -55°C.																				
9 T_C = +25°C	t_{PHL}	3003 Fig. 5	62	2.7 V	2.7 V		2.7 V	OUT		GND	IN	0.0 V	0.0 V	2.7 V	2.7 V	2.7 V	5.0 V	I0 to ΣE	1.5	16.0	ns
			63	"	"		"	"		"	"	2.7 V	0.0 V	"	"	"	"	"			
			64	"	"		"	"		"	"	0.0 V	2.7 V	"	"	"	"	I1 to ΣE			
			65	"	"		"	"		"	"	2.7 V	2.7 V	"	"	"	"	"			
			66	"	"		"	"		"	0.0 V	IN	0.0 V	"	"	"	"	I2 to ΣE			
			67	"	"		"	"		"	2.7 V	"	0.0 V	"	"	"	"	"			
			68	"	"		"	"		"	0.0 V	"	2.7 V	"	"	"	"	I0 to ΣO			
			69	"	"		"	"		"	2.7 V	0.0 V	IN	"	"	"	"	"			
			70	"	"		"	"		"	0.0 V	0.0 V	"	"	"	"	"	I1 to ΣO			
			71	"	"		"	"		"	2.7 V	2.7 V	"	"	"	"	"	"			
			72	"	"		"	"		"	IN	0.0 V	0.0 V	"	"	"	"	I2 to ΣO			
			73	"	"		"	"	OUT	"	"	0.0 V	0.0 V	"	"	"	"	"			
			74	"	"		"	"	"	"	0.0 V	2.7 V	2.7 V	"	"	"	"	I3 to ΣE			
			75	"	"		"	"	"	"	2.7 V	0.0 V	0.0 V	"	"	"	"	"			
			76	"	"		"	"	"	"	0.0 V	2.7 V	2.7 V	"	"	"	"	I4 to ΣE			
			77	"	"		"	"	"	"	2.7 V	IN	0.0 V	"	"	"	"	"			
			78	"	"		"	"	"	"	IN	0.0 V	0.0 V	"	"	"	"	I5 to 1E			
			79	"	"		"	"	"	"	"	2.7 V	2.7 V	"	"	"	"	"			
			80	"	"		"	"	"	"	0.0 V	0.0 V	IN	"	"	"	"	I3 to ΣO			
			81	"	"		"	"	"	"	2.7 V	"	"	"	"	"	"	"			
			82	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			83	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			84	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			85	"	"		"	OUT	"	"	"	"	"	"	"	"	"	"			
			86	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			87	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			88	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			89	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			90	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			91	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			92	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			93	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			94	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			95	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			96	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			97	"	"		"	"	OUT	"	"	"	"	"	"	"	"	"			
			98	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			99	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			100	"	"		"	"	"	"	"	"	"	"	"	"	"	"			
			101	"	"		"	"	"	"	"	"	"	"	"	"	"	"			

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D Case 1/ 2 Test no.	1 2 I6	2 3 I7	3 4 NC	4 6 I8	5 8 ΣE	6 9 ΣO	7 10 GND	8 12 I0	9 13 I1	10 14 I2	11 16 I3	12 18 I4	13 19 I5	14 20 V_{CC}	Measured terminal	Limits Min	Limits Max	Unit
9 T_C = +25°C	t_{PHL}	3003 Fig. 5	102	2.7 V	2.7 V		2.7 V		OUT	GND	2.7 V	2.7 V	2.7 V	0.0 V	IN	0.0 V	5.0 V	I4 to ΣO	1.5	16.0	ns
			103	"	"		"		"	"	"	"	"	2.7 V	"	0.0 V	"	"	"	"	"
			104	"	"		"		"	"	"	"	"	0.0 V	"	2.7 V	"	"	"	"	"
			105	"	"		"		"	"	"	"	"	2.7 V	"	2.7 V	"	"	"	"	"
			106	"	"		"		"	"	"	"	"	0.0 V	0.0 V	IN	"	I5 to ΣO	"	"	"
			107	"	"		"		"	"	"	"	"	2.7 V	0.0 V	"	"	"	"	"	"
			108	"	"		"		"	"	"	"	"	0.0 V	2.7 V	"	"	"	"	"	"
			109	"	"		"		"	"	"	"	"	2.7 V	"	"	"	"	"	"	"
			110	IN	0.0 V		0.0 V	OUT		"	"	"	"	"	"	2.7 V	"	I6 to ΣE	"	"	"
			111	"	2.7 V		0.0 V	"		"	"	"	"	"	"	"	"	"	"	"	"
			112	"	0.0 V		2.7 V	"		"	"	"	"	"	"	"	"	"	"	"	"
			113	"	2.7 V		2.7 V	"		"	"	"	"	"	"	"	"	"	"	"	"
			114	0.0 V	IN		0.0 V	"		"	"	"	"	"	"	"	"	I7 to ΣE	"	"	"
			115	2.7 V	"		0.0 V	"		"	"	"	"	"	"	"	"	"	"	"	"
			116	0.0 V	"		2.7 V	"		"	"	"	"	"	"	"	"	"	"	"	"
			117	2.7 V	"		2.7 V	"		"	"	"	"	"	"	"	"	"	"	"	"
			118	0.0 V	0.0 V		IN	"		"	"	"	"	"	"	"	"	I8 to ΣE	"	"	"
			119	2.7 V	0.0 V		"	"		"	"	"	"	"	"	"	"	"	"	"	"
			120	0.0 V	2.7 V		"	"		"	"	"	"	"	"	"	"	"	"	"	"
			121	2.7 V	2.7 V		"	"		"	"	"	"	"	"	"	"	"	"	"	"
			122	IN	0.0 V		0.0 V		OUT	"	"	"	"	"	"	"	"	I6 to ΣO	"	"	"
			123	"	2.7 V		0.0 V		"	"	"	"	"	"	"	"	"	"	"	"	"
			124	"	0.0 V		2.7 V		"	"	"	"	"	"	"	"	"	"	"	"	"
			125	"	2.7 V		2.7 V		"	"	"	"	"	"	"	"	"	"	"	"	"
			126	0.0 V	IN		0.0 V		"	"	"	"	"	"	"	"	"	I7 to ΣO	"	"	"
			127	2.7 V	"		0.0 V		"	"	"	"	"	"	"	"	"	"	"	"	"
			128	0.0 V	"		2.7 V		"	"	"	"	"	"	"	"	"	"	"	"	"
			129	2.7 V	"		2.7 V		"	"	"	"	"	"	"	"	"	"	"	"	"
			130	0.0 V	0.0 V		IN		"	"	"	"	"	"	"	"	"	I8 to ΣO		"	"
			131	2.7 V	0.0 V		"		"	"	"	"	"	"	"	"	"	"		"	"
			132	0.0 V	2.7 V		"		"	"	"	"	"	"	"	"	"	"		"	"
			133	2.7 V	"		"		"	"	"	"	"	"	"	"	"	"		"	"
	t_{PLH}		134	"	"		2.7 V	OUT		"	IN	0.0 V	0.0 V	"	"	"	"	I0 to ΣE	1.5	15.0	"
			135	"	"		"	"		"	"	2.7 V	0.0 V	"	"	"	"	"	"	"	"
			136	"	"		"	"		"	"	0.0 V	2.7 V	"	"	"	"	"	"	"	"
			137	"	"		"	"		"	"	2.7 V	2.7 V	"	"	"	"	"	"	"	"
			138	"	"		"	"		"	0.0 V	IN	0.0 V	"	"	"	"	I1 to ΣE	"	"	"
			139	"	"		"	"		"	2.7 V	"	0.0 V	"	"	"	"	"	"	"	"
			140	"	"		"	"		"	0.0 V	"	2.7 V	"	"	"	"	"	"	"	"
			141	"	"		"	"		"	2.7 V	"	2.7 V	"	"	"	"	"	"	"	"
			142	"	"		"	"		"	0.0 V	0.0 V	IN	"	"	"	"	I2 to ΣE	"	"	"
			143	"	"		"	"		"	2.7 V	0.0 V	"	"	"	"	"	"	"	"	"
			144	"	"		"	"		"	0.0 V	2.7 V	"	"	"	"	"	"	"	"	"
			145	"	"		"	"		"	2.7 V	"	"	"	"	"	"	"	"	"	"
			146	"	"		"		OUT	"	IN	0.0 V	0.0 V	"	"	"	"	I0 to ΣO	"	"	"
			147	"	"		"		"	"	"	2.7 V	0.0 V	"	"	"	"	"	"	"	"
			148	"	"		"		"	"	"	0.0 V	2.7 V	"	"	"	"	"	"	"	"
			149	"	"		"		"	"	"	2.7 V	2.7 V	"	"	"	"	"	"	"	"
			150	"	"		"		"	"	0.0 V	IN	0.0 V	"	"	"	"	I1 to ΣO	"	"	"
			151	"	"		"		"	"	2.7 V	"	0.0 V	"	"	"	"	"	"	"	"
			152	"	"		"		"	"	0.0 V	"	2.7 V	"	"	"	"	"	"	"	"
			153	"	"		"		"	"	2.7 V	"	2.7 V	"	"	"	"	"	"	"	"
			154	"	"		"		"	"	0.0 V	0.0 V	IN	"	"	"	"	I2 to ΣO	"	"	"
			155	"	"		"		"	"	2.7 V	0.0 V	"	"	"	"	"	"	"	"	"
			156	"	"		"		"	"	0.0 V	2.7 V	"	"	"	"	"	"	"	"	"
			157	"	"		"		"	"	2.7 V	"	"	"	"	"	"	"	"	"	"
			158	"	"		"	OUT		"	"	"	2.7 V	IN	0.0 V	0.0 V	"	I3 to ΣE	"	"	"
			159	"	"		"	"		"	"	"	"	"	2.7 V	0.0 V	"	"	"	"	"
			160	"	"		"	"		"	"	"	"	"	0.0 V	2.7 V	"	"	"	"	"
			161	"	"		"	"		"	"	"	"	"	2.7 V	2.7 V	"	"	"	"	"

See footnotes at end of table.

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TABLE III. Group A inspection for device type 01 - Continued.
Terminal conditions (pins not designated may be high ≥2.0 V, low ≤0.8 V, or open)

Subgroup	Symbol	MIL-STD-883 method	Cases A,B,C,D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	Measured terminal	Limits		Unit
			Case 1/ 2	2	3	4	6	8	9	10	12	13	14	16	18	19	20		Min	Max	
			Test no.	16	17	NC	18	ΣE	ΣO	GND	10	11	12	13	14	15	V_{CC}				
9 T_C = +25°C	t_{PLH}	3003 Fig. 5	162	2.7 V	2.7 V		2.7 V	OUT		GND	2.7 V	2.7 V	2.7 V	0.0 V	IN	0.0 V	5.0 V	I4 to ΣE	1.5	15.0	ns
			163	"	"		"	"		"	"	"	"	2.7 V	"	0.0 V	"	"	"	"	"
			164	"	"		"	"		"	"	"	"	0.0 V	"	2.7 V	"	"	"	"	"
			165	"	"		"	"		"	"	"	"	2.7 V	"	2.7 V	"		"	"	"
			166	"	"		"	"		"	"	"	"	0.0 V	0.0 V	IN	"	I5 to IΣ	"	"	"
			167	"	"		"	"		"	"	"	"	2.7 V	0.0 V	"	"		"	"	"
			168	"	"		"	"		"	"	"	"	0.0 V	2.7 V	"	"		"	"	"
			169	"	"		"	"		"	"	"	"	2.7 V	2.7 V	"	"		"	"	"
			170	"	"		"	"	OUT	"	"	"	"	IN	0.0 V	0.0 V	"	I3 to ΣO	"	"	"
			171	"	"		"	"	"	"	"	"	"	"	2.7 V	0.0 V	"		"	"	"
			172	"	"		"	"	"	"	"	"	"	"	0.0 V	2.7 V	"		"	"	"
			173	"	"		"	"	"	"	"	"	"	0.0 V	2.7 V	0.0 V	"	I4 to ΣO	"	"	"
			174	"	"		"	"	"	"	"	"	"	2.7 V	0.0 V	2.7 V	"		"	"	"
			175	"	"		"	"	"	"	"	"	"	"	2.7 V	0.0 V	"		"	"	"
			176	"	"		"	"	"	"	"	"	"	"	0.0 V	2.7 V	"	I5 to ΣO	"	"	"
			177	"	"		"	"	"	"	"	"	"	"	2.7 V	0.0 V	"		"	"	"
			178	"	"		"	"	"	"	"	"	"	"	0.0 V	2.7 V	"		"	"	"
			179	"	"		"	"	"	"	"	"	"	"	2.7 V	0.0 V	"	I6 to ΣE	"	"	"
			180	"	"		"	"	"	"	"	"	"	"	"	2.7 V	"		"	"	"
			181	"	"		"	"	"	"	"	"	"	"	"	"	"	I7 to ΣE	"	"	"
			182	IN	0.0 V		0.0 V	OUT	"	"	"	"	"	"	"	"	"		"	"	"
			183	"	2.7 V		0.0 V	"	"	"	"	"	"	"	"	"	"	I8 to ΣE	"	"	"
			184	"	0.0 V		2.7 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			185	"	2.7 V		2.7 V	"	"	"	"	"	"	"	"	"	"	I6 to ΣO	"	"	"
			186	0.0 V	IN		0.0 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			187	2.7 V	"		0.0 V	"	"	"	"	"	"	"	"	"	"	I7 to ΣO	"	"	"
			188	0.0 V	"		2.7 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			189	2.7 V	"		2.7 V	"	"	"	"	"	"	"	"	"	"	I8 to ΣO	"	"	"
			190	0.0 V	0.0 V		IN	"	"	"	"	"	"	"	"	"	"		"	"	"
			191	2.7 V	0.0 V		"	"	"	"	"	"	"	"	"	"	"		"	"	"
			192	0.0 V	2.7 V		"	"	"	"	"	"	"	"	"	"	"		"	"	"
			193	2.7 V	2.7 V		"	"	"	"	"	"	"	"	"	"	"		"	"	"
			194	IN	0.0 V		0.0 V	"	OUT	"	"	"	"	"	"	"	"		"	"	"
			195	"	2.7 V		0.0 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			196	"	0.0 V		2.7 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			197	"	2.7 V		2.7 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			198	0.0 V	IN		0.0 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			199	2.7 V	"		0.0 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			200	0.0 V	"		2.7 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			201	2.7 V	"		2.7 V	"	"	"	"	"	"	"	"	"	"		"	"	"
			202	0.0 V	0.0 V		IN	"	"	"	"	"	"	"	"	"	"		"	"	"
			203	2.7 V	0.0 V		"	"	"	"	"	"	"	"	"	"	"		"	"	"
			204	0.0 V	2.7 V		"	"	"	"	"	"	"	"	"	"	"		"	"	"
			205	2.7 V	2.7 V		"	"	"	"	"	"	"	"	"	"	"		"	"	"
																			1.0	21.0	"
10	t_{PHL}	Same tests and terminal conditions as for subgroup 9, except T_C = +125°C and limits as shown.																	1.0	20.0	"
	t_{PLH}																				
11		Same tests, terminal conditions, and limits as for subgroup 10, except T_C = -55°C.																			

1/ For case 2, pins not referenced are N/C.

2/ I_{IL} limits shall be as follows:

Test	Min/max limits in mA		
	Circuits		
	A	B	C
I_{IL}	-.25/-.60	-.03/-.60	

3/ Perform function sequence at V_{CC} = 4.5 V and repeat at V_{CC} = 5.5 V.

4/ A = 4.5 V, B = 0.0 V.
H ≥2.5 V, L ≤0.5 V with high speed checker double comparator, or
H ≥1.5 V, L ≤1.5 V with high speed checker single comparator.

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4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, and 6 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883. Electrical parameters shall be as specified in table II herein.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D or E, using the circuit shown on figure 4, or equivalent.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be specified as follows.

4.5.1 Voltage and current. All voltages given are referenced to the microcircuit ground terminal. Currents given are conventional and positive when flowing into the referenced terminal.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.2 Ordering data. The acquisition document should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

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d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements shall not apply to direct purchase by or direct shipment to the Government.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

GND	Ground zero voltage potential.
I_{IN}	Current flowing into an input terminal.
V_{IN}	Voltage level at an input terminal

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	54F280

6.6 Manufacturers' designations. Manufacturers' circuits which form a part of this specification are designated as shown in table IV herein.

TABLE IV. Manufacturers' designations.

Device type	Circuits		
	A	B	C
	Fairchild Semiconductor	Motorola Inc.	Signetics Corp.
01	X	X	

6.7 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI
Navy - OS, SH, TD
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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